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Battery and method of manufacturing same

Abstract

A battery case includes a case-penetrating hole. An external terminal includes a member-engaging hole and is stacked on an outside of the battery case so that the member-engaging hole and the case-penetrating hole align with each other. An internal terminal includes a shaft portion and a fastening portion. The shaft portion is inserted through the case-penetrating hole and the member-engaging hole from an interior of the battery. The fastening portion is fastened to a circumferential periphery of the member-engaging hole by crushing a tip end of the shaft portion. The internal terminal is stacked on an inside of the battery case. The fastening portion and the external terminal include a welded portion welded along a weld line set across a periphery of the fastening portion. The welded portion has a length K1 along the weld line and a width K2 along a direction perpendicular to the weld line, and the length K1 is at least 1.5 times longer than the width K2.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This is a divisional of application Ser. No. 16/655,941 filed Oct. 17, 2019, which claims priority from Japanese Patent Application No. 2018-197011 filed on Oct. 18, 2018, which is incorporated by reference herein in its entirety.

BACKGROUND

(1) The present disclosure relates to a battery and a method of manufacturing the battery.

(2) PCT/International Publication No. WO 2013/031669 discloses a battery in which an internal terminal is joined by riveting and an outer peripheral portion of the joined part is additionally welded. The publication describes that this enables the joined part between the internal terminal and an external terminal to provide increased reliability and reduce the internal resistance.

SUMMARY

(3) However, when the internal terminal and the external terminal are welded together by spot welding, as described in WO 2013/031669, the thickness of the weld spot will vary from one location to another. This means that when the weld area is small, the electrical resistance in the joined part increases, resulting in a higher battery internal resistance. For this reason, it is necessary to provide a large weld area in order to reduce the electrical resistance in the joined part and to thereby lower the battery internal resistance.

(4) A battery according to the present disclosure includes a battery case, an external terminal, and an internal terminal. The battery case includes a case-penetrating hole. The external terminal includes a member-engaging hole and is stacked on an outside of the battery case so that the member-engaging hole and the case-penetrating hole align with each other. The internal terminal is stacked on an inside of the battery case, and includes a shaft portion and a fastening portion. The shaft portion is inserted through the case-penetrating hole and the member-engaging hole from an interior of the battery. The fastening portion is fastened to a circumferential periphery of the member-engaging hole at a tip end of the shaft portion. The fastening portion and the external terminal include a welded portion welded along a weld line set across a periphery of the fastening portion. The welded portion has a length K1 along the weld line and a width K2 along a direction perpendicular to the weld line, and the length K1 is at least 1.5 times longer than the width K2. The above-described embodiment of the battery makes it possible to reduce the internal resistance of the battery.

(5) It is possible that a portion of the weld line that is provided external to the fastening portion may be at least 1.5 times longer than a portion of the weld line that is provided internal to the fastening portion. With the above-described embodiment of the battery, the weld area can be increased, and therefore, the internal resistance of the battery can be reduced.

(6) Herein, the at least one welded portion may include at least two or more of the welded portions, and adjacent ones of the welded portions may be circumferentially spaced from each other. The above-described embodiment of the battery makes it possible to further reduce the electrical resistance.

(7) Herein, the welded portion may include an inner edge of the fastening portion, and the inner edge may be located radially outward of the member-engaging hole when viewed axially along the shaft portion. With the above-described embodiment of the battery, the fastened portion is unlikely to be adversely affected by welding.

(8) According to the present disclosure, an embodiment of the method of manufacturing a battery includes the steps of: preparing a battery case including a case-penetrating hole; preparing an external terminal including a member-engaging hole; preparing an internal terminal including a

shaft portion; assembling the battery case, the external terminal, and the internal terminal together so that the external terminal is stacked on an outside of the battery case, the internal terminal is stacked on an inside of the battery case, and the shaft portion is inserted through the case-penetrating hole and the member-engaging hole; forming a fastening portion by crushing a tip end of the shaft portion and fastening the tip end of the shaft portion to a circumferential periphery of the member-engaging hole on an outside of the external terminal; and laser welding the fastening portion and the external terminal seamlessly along at least one weld line that is set so as to extend across a periphery of the fastening portion along an outer surface of the fastening portion. The above-described embodiment of the method of manufacturing a battery makes it possible to reduce the internal resistance of the battery.

(9) Herein, in the step of laser welding, a region external to the fastening portion may be laser welded at least 2.5 times longer than a region internal to the fastening portion along the at least one weld line. The above-described embodiment of the method of manufacturing a battery serves to prevent the external terminal from cracking.

(10) Herein, in the step of laser welding, the at least one weld line may be set at two or more locations of the fastening portion that are circumferentially spaced from each other, the fastening portion and the external terminal may be laser welded together along a plurality of the weld lines, and a plurality of welded portions that are circumferentially spaced from each other may be formed by the laser welding. The above-described embodiment of the method of manufacturing a battery serves to reduce the weld defects resulting from the occurrence of spatter.

(11) Herein, in the step of laser welding, the welded portion may not extend into an inside of the member-engaging hole when viewed axially along the shaft portion. With the above-described embodiment of the method of manufacturing a battery, the fastened portion is unlikely to be adversely affected by welding.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a partial cross-sectional view of a battery according to an embodiment of the present disclosure.
- (2) FIG. 2 is a cross-sectional view illustrating a portion where an external terminal and an internal terminal are fitted to a battery case, according to an embodiment of the present disclosure.
- (3) FIG. 3 is a partial plan view illustrating a welded portion formed in a fastening portion of the internal terminal and the external terminal, according to an embodiment of the present disclosure.
- (4) FIG. 4 is a cross-sectional view taken along line IV-IV in FIG. 3, showing a fastening portion **15c1**.
- (5) FIG. 5 is a partial plan view illustrating a welded portion formed in the fastening portion of the internal terminal and the external terminal, according to a modified embodiment of the present disclosure.

DETAILED DESCRIPTION

(6) Hereinbelow, preferred embodiments of the present invention will be described. It should be noted that the following embodiments are merely illustrative of the present invention and the present invention is not limited to the following embodiments. The drawings are schematic illustrations, and do not necessarily reflect any actual product.

(7) FIG. 1 is a partial cross-sectional view of a battery **10** according to one embodiment of the present disclosure. FIG. 1 depicts the interior of the battery **10** that is exposed along one wide side surface of a battery case **11** in substantially a rectangular parallelepiped shape. FIG. 2 is a cross-sectional view illustrating a portion where an external terminal **14** and an internal terminal **15** are fitted to a battery case **11**. As illustrated in FIGS. 1 and 2, the battery **10** includes the battery case

11, insulating members **12** and **13**, external terminals **14**, internal terminals **15**, and an electrode assembly **16**.

(8) Here, as illustrated in FIG. **1**, the external terminals **14** and the internal terminals **15** are fitted to the battery case **11** with the insulating members **12** and **13** interposed therebetween. The battery case **11** encloses the electrode assembly **16** and an electrolyte solution, which is not shown. The electrode assembly **16** is covered with an insulating film or the like, which is not shown, and is enclosed in the battery case **11**. Thus, the battery case **11** and the electrode assembly **16** are insulated from each other. The electrode assembly **16** includes a positive electrode element, a negative electrode element, and a separator of the battery **10**.

(9) In the present embodiment, a long sheet-shaped positive electrode sheet **21** and a long sheet-shaped negative electrode sheet **22** are stacked with two long sheet-shaped separator sheets **31** and **32** alternately interposed between them, and the positive electrode sheet **21**, the negative electrode sheet **22**, and the separator sheets **31** and **32** are wound together to form the electrode assembly **16**.

(10) The positive electrode sheet **21** includes a positive electrode current collector foil **21a** having a predetermined width and a predetermined thickness, a positive electrode active material layer **21b** containing a positive electrode active material, and an uncoated portion **21a1** defined along one lateral edge of the positive electrode current collector foil **21a** with a constant width. The positive electrode active material layer **21b** is formed on both faces of the positive electrode current collector foil **21a**, except for the uncoated portion **21a1**. In a lithium-ion secondary battery, for example, the positive electrode active material is a material that is capable of releasing lithium ions during charge and absorbing lithium ions during discharge, such as a lithium-transition metal composite oxide. Other than the lithium-transition metal composite oxide, various materials have been proposed for use as the positive electrode active material, and the positive electrode active material is not limited to a particular material.

(11) The negative electrode sheet **22** includes a negative electrode current collector foil **22a** having a predetermined width and a predetermined thickness, a negative electrode active material layer **22b** containing a negative electrode active material, and an uncoated portion **22a1** defined along one lateral edge of the negative electrode current collector foil **22a** with a constant width. The negative electrode active material layer **22b** is formed on both faces of the negative electrode current collector foil **22a**, except for the uncoated portion **22a1**. In a lithium-ion secondary battery, for example, the negative electrode active material is a material that is capable of absorbing lithium ions during charge and releasing the absorbed lithium ions during discharge, such as graphite. Other than graphite, various materials have been proposed for use as the negative electrode active material, and the negative electrode active material is not limited to a particular material.

(12) Each of the separator sheets **31** and **32** may be formed of, for example, an electrolyte permeable porous resin sheet that has required heat resistance. Various proposals have been made about the separator sheets **31** and **32**, and there is no particular restriction on the separator sheets **31** and **32**.

(13) The electrode assembly **16** is deformed into a flat shape pressed in one direction perpendicular to the winding axis and is enclosed in the battery case **11**. Herein, the battery case **11** includes a case main body **11a** and a lid **11b**. The case main body **11a** has a flat prismatic housing region, and one side end thereof is open. The lid **11b** has a shape corresponding to the opening of the case main body **11a**.

(14) In the electrode assembly **16**, the uncoated portion **21a1** of the positive electrode sheet **21** and the uncoated portion **22a1** of the negative electrode sheet **22** protrude laterally in opposite directions from the separator sheets **31** and **32**. The uncoated portion **21a1** of the positive electrode current collector foil **21a** and the uncoated portion **22a1** of the negative electrode current collector foil **22a** are connected to respective internal terminals **15**, which are respectively attached to opposite side portions of the battery case **11**. The electrode assembly **16** is enclosed in the flat internal space of the case main body **11a**. After the electrode assembly **16** is placed therein, the case

main body **11a** is closed by the lid **11b**.

(15) In the present embodiment, as illustrated in FIG. 1, the external terminal **14** and the internal terminal **15** are fitted to each of longitudinally opposite side end portions of the lid **11b**. Herein, each of the internal terminals **15** includes a base portion **15a**, an attachment piece **15b**, and a shaft portion **15c**. The base portion **15a** is fitted to the lid **11b** with an insulating member **12** interposed therebetween. The attachment piece **15b** extends from the base portion **15a** into the interior of the battery case **11**. Referring to FIG. 1, the attachment piece **15b** of the internal terminal **15** that is shown on the left side of the figure is welded to the uncoated portion **21a1** of the positive electrode current collector foil **21a** of the electrode assembly **16**. The attachment piece **15b** of the internal terminal **15** that is shown on the right side of the figure is welded to the uncoated portion **22a1** of the negative electrode current collector foil **22a** of the electrode assembly **16**. On the outside of the lid **11b**, the external terminal **14** that is shown on the left side is fitted to a positive electrode connecting terminal **41**. The external terminal **14** that is shown on the right side is fitted to a negative electrode connecting terminal **42**.

(16) As illustrated in FIG. 2, the lid **11b** includes a case-penetrating hole **11c** for mounting the external terminal **14** and the internal terminal **15**. In addition, a recess **11b1** is formed in an outer surface of the lid **11b** at a position at which an insulating member **13** for insulating the external terminal **14** is to be fitted. The same structure is also provided on the negative electrode side of the lid **11** to which the external terminal **14** and the internal terminal **15** are fitted. On the positive electrode side, the external terminal **14**, the internal terminal **15**, and the connecting terminal **41** are made of aluminum or an aluminum alloy. On the negative electrode side, the external terminal **14**, the internal terminal **15**, and the connecting terminal **42** are made of copper or a copper alloy.

(17) The insulating members **12** and **13** are fitted around the case-penetrating hole **11c**.

(18) The first insulating member **12** is disposed on an inside of the lid **11b** to provide electrical insulation between the lid **11b** and the internal terminal **15**. In the present embodiment, the first insulating member **12** is composed of a resin member having required elasticity. Specifically, the first insulating member **12** is composed of a fluoropolymer. The first insulating member **12** ensures sufficient sealing of the case-penetrating hole **11c**. In the present embodiment, as illustrated in FIG. 2, the first insulating member **12** includes a base portion **12a**, a tubular portion **12b**, and a retaining portion **12c**. The base portion **12a** is a portion to be fitted on an inner surface of the lid **11b**. The tubular portion **12b** is a portion of the first insulating member **12** that protrudes from the base portion **12a** and is to be fitted onto the inner circumferential surface of the case-penetrating hole **11c**. The tubular portion **12b** has an internal diameter corresponding to the outer diameter of the shaft portion **15c** so that a later-described shaft portion **15c** of the internal terminal **15** can be inserted therethrough. The retaining portion **12c** is provided to be the lower surface of the first insulating member **12**. The retaining portion **12c** includes a recess that corresponds to the shape of a later-described base portion **15a** of the internal terminal **15**. The retaining portion **12c** is a portion of the first insulating member **12** on which the base portion **15a** is to be positioned.

(19) The second insulating member **13** is disposed on an outside of the lid **11b** to provide electrical insulation between the lid **11b** and the external terminal **14**, and electrical insulation between the lid **11b** and the connecting terminal **41**. In the present embodiment, the second insulating member **13** is composed of a resin member. Specifically, the second insulating member **13** is composed of a polyamide resin. In this embodiment, the lower surface of the second insulating member **13** is provided with a protruding portion **13a**, as illustrated in FIG. 2. The protruding portion **13a** is fitted into the recess **11b1** of the lid **11b**. The upper surface of the second insulating member **13** is provided with a recess **13b**, in which the positive electrode connecting terminal **41** is to be disposed, and a recess **13c**, in which the external terminal **14** is to be fitted. The recess **13c**, in which the external terminal **14** is to be fitted, includes a through hole **13d** formed at a position that corresponds to the case-penetrating hole **11c** formed in the lid **11b**. The through hole **13d** has an internal diameter corresponding to the outer diameter of the shaft portion **15c** so that a later-

described shaft portion **15c** of the internal terminal **15** can be inserted therethrough.

(20) In the present embodiment, the insulating members **12** and **13** are composed of two component parts. However, the insulating members **12** and **13** are component parts that provides electrical insulation between the lid **11b**, the external terminal **14**, and the internal terminal **15**, and it should be noted that the materials, the shapes, and the structures mentioned above are merely examples. The insulating members **12** and **13** may be composed of a single component part. Alternatively, the insulating members **12** and **13** may be composed of a plurality of component parts, such as two or more component parts.

(21) The connecting terminal **41** includes a flange portion **41a** and a shaft portion **41b**. The flange portion **41a** is positioned and fitted into the recess **13b**, which is provided in the second insulating member **13**, which is disposed on the outside of the lid **11b**. Accordingly, the flange portion **41a** and the recess **13b** may have shapes corresponding to each other. Note that the shaft portion **41b** is a part that serves as an output terminal.

(22) In the present embodiment, as illustrated in FIG. 2, the external terminal **14** includes a member-engaging hole **14b** and is stacked on an outside of the lid **11b** so that the member-engaging hole **14b** and the case-penetrating hole **11c** align with each other. The external terminal **14** is a plate-shaped member disposed along the longitudinal axis of the lid **11b**. The external terminal **14** includes an attachment hole **14a**, to which the connecting terminal **41** is attached, and a member-engaging hole **14b**, through which the later-described shaft portion **15c** of the internal terminal **15** is inserted. The member-engaging hole **14b** is formed at a position corresponding to the through hole **13d** of the second insulating member **13** under conditions where the external terminal **14** is placed on the second insulating member **13**. Thus, in the present embodiment, the external terminal **14** is stacked on the outside of the lid **11b** with the second insulating member **13** interposed therebetween, because the lid **11b** is an electrically conductive member.

(23) In the present embodiment, the internal terminal **15** includes a shaft portion **15c** and a fastening portion **15c1** and is stacked on the inside of the lid **11b**. The shaft portion **15c** is inserted through the case-penetrating hole **11c** and the member-engaging hole **14b** from the interior of the battery case **11**. The fastening portion **15c1** is a portion of the shaft portion **15c** that is fastened to a circumferential periphery of the member-engaging hole **14b** by crushing a tip end of the shaft portion **15c**. As illustrated in FIGS. 1 and 2, the internal terminal **15** includes a base portion **15a**, an attachment piece **15b**, and a shaft portion **15c**. Of these, the base portion **15a** is fitted to the lid **11b** with the first insulating member **12** interposed therebetween. The details of the attachment piece **15b** have already been described and therefore will not be repeated herein.

(24) FIG. 3 is a partial plan view illustrating welded portions **50** formed in the fastening portion **15c1** of the internal terminal **15** and the external terminal **14**, according to the present embodiment. In FIG. 3, the welded portions are enlarged to facilitate understanding. In the present embodiment, as illustrated in FIGS. 2 and 3, the fastening portion **15c1** and the external terminal **15** include welded portions **50** that are seamlessly welded by using laser light along a weld line W that is set across the periphery of the fastening portion **15c1**. Herein, the term “weld line W” refers to a scanning line of the laser beam in laser welding. The weld line W may appear as a line through the center of the weld width of the welded portion **50**, for example. In the present embodiment, each of the welded portions **50** has a length K1 along the weld line W and a width K2 along a direction perpendicular to the weld line W, and the length K1 is greater than the width K2. In the present embodiment, the length K1 is set to be approximately from 1.5 times to 4 times the width K2. As a result, the welded portion **50** can be distinguished from one that is formed by simply spot welding the boundary portion between the fastening portion **15c1** and the external terminal **14**.

(25) In the present embodiment, the weld is formed along the weld line W that is set across the periphery of the fastening portion **15c1**. Therefore, each of the welded portions **50** is formed across the periphery of the fastening portion **15c1**. In this case, the external terminal **14** is sufficiently fused not only in an inside part of the fastening portion **15c1** but also in an outside part of the

fastening portion **15c1** so that the fastening portion **15c1** and the external terminal **14** are welded to each other. More specifically, the metal in the fastening portion **15c1** that is fused by laser welding flows along the outer surface of the fastening portion **15c1**, which is in an umbrella-like shape, to the external terminal **14**, and is mixed with the fused metal on the external terminal **14** side, to effect welding. As a result, the weld area can be made larger than, for example, the case where the boundary portion between the fastening portion **15c1** and the external terminal **14** is simply spot welded. Moreover, electrical continuity between the external terminal **14** and the fastening portion **15c1** of the internal terminal **15** can be ensured more reliably, and moreover, the electrical resistance between them can be further reduced. Therefore, the weld area can be made larger in comparison with the case of spot welding, and as a result, the internal resistance of the battery can be reduced. In order to obtain such an advantageous effect, the welded portion **50** may be such that the length $K1$ along the weld line W is greater than the width $K2$ along a direction perpendicular to the weld line W . For example, the length $K1$ may be at least 1.5 times, more preferably at least 2 times, still more preferably at least 2.5 times the width $K2$. Such a welded portion may be embodied by, for example, performing laser welding while shifting the position to which laser is applied along the weld line W .

(26) FIG. 4 is a cross-sectional view taken along line IV-IV in FIG. 3, showing a fastening portion **15c1**. The shaft portion **15c** of the internal terminal **15** is inserted through the case-penetrating hole **11c** of the lid **11b** and the member-engaging hole **14b** of the external terminal **14**. The tip end of the shaft portion **15c** is crushed and fastened to a circumferential periphery of the member-engaging hole **14b** that is on the outside of the external terminal **14**. The tip end of the shaft portion **15c** is deformed so as to spread into an umbrella-like shape to form the fastening portion **15c1**. As a result, the internal terminal **15** is secured so as to clamp the first insulating member **12**, the lid **11b**, the second insulating member **13**, and the external terminal **14** between the base portion **15a** and the fastening portion **15c1**.

(27) Of the weld line W , a weld line $L_{sub.out}$ that is provided external to the fastening portion **15c1** may be set longer than a weld line $L_{sub.in}$ that is provided internal to the fastening portion **15c1**. For example, the weld line $L_{sub.out}$, which is provided external to the fastening portion **15c1**, may be set to be at least 2.5 times longer than the weld line $L_{sub.in}$, which is provided internal to the fastening portion **15c1**. In other words, a portion of the welded portion **50** that is provided external to the fastening portion **15c1** may have a length along the weld line W at least 2.5 times longer than a portion of the welded portion **50** that is provided internal to the fastening portion **15c1**. Here, the distance from the center of the shaft portion **15c** of the internal terminal **15** to the outer periphery of the fastening portion **15c1** is defined as the distance r from the center of the fastening portion **15c1** to the outer periphery of the fastening portion **15c1**. The weld line $L_{sub.out}$, which is provided external to the fastening portion **15c1**, may be at least 0.25 times, preferably at least 0.3 times, the distance R from the center of the fastening portion **15c1** to the outer periphery thereof. On the other hand, the weld line $L_{sub.in}$, which is provided internal to the fastening portion **15c1**, may be at least 0.1 times, preferably at least 0.15 times, the distance R from the center of the fastening portion **15c1** to the outer periphery thereof.

(28) It should be noted that, when the fastening portion **15c1** is in a substantially circular shape centered about the center of the shaft portion **15c** of the internal terminal **15**, the distance r from the center of the fastening portion **15c1** to the outer periphery thereof can be approximately the radius of the fastening portion **15c1**. Thus, because the weld line $L_{sub.out}$ provided external to the fastening portion **15c1** is set longer than the weld line $L_{sub.in}$ provided internal to the fastening portion **15c1**, the weld area is increased further, and the electrical resistance is reduced. Also, because the metal on the external terminal **14** side is fused in a greater amount in a region external to the fastening portion **15c1**, the thermal strain that may occur in the external terminal **14** is reduced, so that the quality of the external terminal **14** can be kept high.

(29) As illustrated in FIG. 4, the welded portion **50** may be provided at only one location. It is also

possible that two or more of the welded portions **50** may be provided at respective locations. In that case, it is desirable that the welded portions **50** be circumferentially spaced from each other. With the numbers of the welded portions being equal, such a configuration provides a larger weld area than the case where the welded portions **50** are circumferentially overlapped with each other, and therefore allows the electrical resistance to be kept even lower.

(30) The region in which the external terminal **14** and the shaft portion **15c** of the internal terminal **15** are in contact with each other is under the force exerted by fastening. In the present embodiment, the inner edge of the fastening portion **15c1** within the welded portion may be located radially outward of the member-engaging hole **14b** when viewed axially along the shaft portion **15c**. With such a configuration, welding is unlikely to adversely affect the region that is fastened by the fastening force. As a result, the joining of the external terminal **14** and the internal terminal **15** that is effected by fastening can be kept firmly. From such viewpoints, the inner edge of the fastening portion of the welded portion is set to be located at least 1.1 times, more preferably at least 1.2 times, further outward than the radius of the member-engaging hole when viewed axially along the shaft portion.

(31) In the foregoing embodiment, the weld line **W** extends in a radial direction, and the center of the shaft portion **15c** is on an extension line of the weld line **W**. However, the present disclosure is not limited thereto. FIG. 5 is a partial plan view illustrating welded portions **50** formed in the external terminal and the fastening portion of the internal terminal, according to a modified embodiment of the present disclosure. As illustrated in FIG. 5, the weld line **W** may be set to be offset from a line along a radial direction through the center of the shaft portion **15c** in a direction perpendicular to the radial direction. In addition, the weld line **W** is not limited to a straight line but may be set to be a curved line.

(32) In the foregoing embodiment, the welded portions **50** are provided symmetrically. However, the present disclosure is not limited thereto. As illustrated in FIG. 5, the numbers and the locations of the welded portions **50** may be asymmetrical.

(33) In the foregoing embodiment, the external terminal **14** and the internal terminal **15** are provided on the lid **11b**. However, the present disclosure is not limited thereto, and the external terminal **14** and the internal terminal **15** may be provided on the battery case **11**.

(34) The external terminal **14** and the internal terminal **15** may be made of dissimilar materials. For example, in secondary batteries that have a charge potential such that the open circuit voltage of the positive electrode and the negative electrode is about 4 V, the electrode terminal on the positive electrode uses aluminum or an aluminum alloy that can withstand a required voltage. In such cases, dissimilar materials may be used for the external terminal **14** and the internal terminal **15**. For example, in the embodiment shown in FIGS. 2 and 3, the internal terminal **15** of the positive electrode that includes the shaft portion **15c**, which needs to be crushed, may use aluminum, which is relatively a soft material. On the other hand, the external terminal **14** may use an aluminum alloy that shows excellent corrosion resistance and strength.

(35) A method of manufacturing a battery that includes such a welded portion **50** will be described below. In the method of manufacturing a battery herein, a battery case **11** including a case-penetrating hole **11c**, an external terminal **14** including a member-engaging hole **14b**, and an internal terminal **15** including a shaft portion **15c** are prepared.

(36) Next, the battery case **11**, the external terminal **14**, and the internal terminal **15** are assembled together. In this step, the external terminal **14** is placed on an outside of the battery case **11**, the internal terminal **15** is placed on an inside of the battery case **11**, and the shaft portion **15c** is inserted through the case-penetrating hole **11c** and the member-engaging hole **14b**. In the preferred embodiment, the first insulating member **12** is internal terminal between the battery case **11** and the internal terminal **15**. The second insulating member **13** is interposed between the battery case **11** and the external terminal **14**.

(37) Specifically, the external terminal **14** is placed on the outside of the lid **11b** so that the second

insulating member **13** is interposed therebetween and also the positions of the case-penetrating hole **11c** and the member-engaging hole **14b** are aligned with each other. Next, the internal terminal **15** is placed on the inside of the lid **11b** so that the first insulating member **12** is interposed therebetween and the shaft portion **15c** of the internal terminal **15** is inserted through the case-penetrating hole **11c** and the member-engaging hole **14b**.

(38) Subsequently, a tip end of the shaft portion **15c** is crushed and fastened to the circumferential periphery of the member-engaging hole **14b** that is on the outside of the external terminal **14**, to form the fastening portion **15c1**. Thus, the internal terminal **15** is secured so as to clamp the first insulating member **12**, the lid **11b**, the second insulating member **13**, and the external terminal **14** between the base portion **15a** and the fastening portion **15c1**.

(39) Then, the fastening portion **15c1** and the external terminal **14** are seamlessly laser welded along the weld line W that is set so as to extend across the periphery of the fastening portion **15c1** along the outer surface of the fastening portion **15c1**. In such a configuration, the fastening portion **15c1** is in an umbrella-like shape. Therefore, the metal of the fastening portion **15c1** that is fused by the laser light flows toward the external terminal **14**. Also, the external terminal **14** is fused by the laser light in a region external to the fastening portion **15c1**, so the metal of the fastening portion **15c1** is mixed and welded with the metal of the external terminal **14**. As a result, the weld area is increased and the electrical resistance is reduced, so that the internal resistance of the battery **10** is reduced.

(40) The welded portion may be provided at either one location or two or more locations. When a plurality of locations are to be welded by laser light, a plurality of welded portions may be provided so as to be spaced from each other along a circumferential direction of the fastening portion **15c1**. In that case, for example, in the step of laser welding the fastening portion **15c1** and the external terminal **14** to each other, weld lines W may be set at two or more locations that are spaced along a circumferential direction of the fastening portion **15c1**, and the fastening portion **15c1** and the external terminal **14** may be laser welded to each other along the plurality of weld lines W. When a plurality of welded portions are formed so as to be spaced from each other along a circumferential direction of the fastening portion **15c1** in this way, the occurrence of spatter resulting from the heat input in welding can be reduced, and accordingly, weld defects can be reduced.

(41) In the step of seamlessly laser welding the fastening portion **15c1** and the external terminal **14**, a region that is to be laser welded may be set to be at least 2.5 times longer in a region external to the fastening portion **15c1** than in a region internal to the fastening portion **15c1**. In this case, the weld line L.sub.out that is provided external to the fastening portion **15c1** may be set at least 2.5 times longer than the weld line L.sub.in that is provided internal to the fastening portion **15c1**. When the external terminal **14** and the fastening portion **15c1** are made of dissimilar materials, their thermal expansion coefficients may be different. When that is the case, as the welded portion **50** fused by the laser light undergoes heat shrinkage according to a temperature drop, a thermal stress resulting from the difference in thermal expansion occurs between the external terminal **14** and the welded portion **50**. When the region that is to be laser welded is set to be at least 2.5 times longer in a portion external to the fastening portion **15c1** than that in a portion internal to the fastening portion **15c1** as described above, it means that the external terminal **14** is fused in a wider area. This serves to lessen the temperature dropping rate of the welded portion **50**. Therefore, even if the welded portion **50** undergoes heat shrinkage, the thermal stress produced in and around the welded portion **50** is kept small.

(42) In addition, in the step of laser welding the fastening portion **15c1** and the external terminal **14** to each other, the fastening portion **15c1** and the external terminal **14** are welded so that the welded portion **50** does not extend into the inside of the member-engaging hole **14** when viewed axially along the shaft portion **15c**. In other words, the inner edge of the fastening portion of the welded portion is set to be located at least 1.1 times further outward than the radius of the member-engaging hole when viewed axially along the shaft portion. With such an embodiment, laser

welding is unlikely to adversely affect the region of the fastening portion **15c1** that is under a fastening force. As a result, the joining of the external terminal **14** and the internal terminal **15** that is effected by fastening is kept firmly.

(43) Although various embodiments of the battery and the method of manufacturing the battery have been described hereinabove according to the present disclosure, it should be noted that, unless specifically stated otherwise, the embodiments and the examples described herein do not limit the scope of the present invention.

Claims

1. A method of manufacturing a battery, comprising the steps of; preparing a battery case including a case-penetrating hole; preparing an external terminal including a member-engaging hole; preparing an internal terminal including a shaft portion; assembling the battery case, the external terminal, and the internal terminal together so that the external terminal is stacked on an outside of the battery case, the internal terminal is stacked on an inside of the battery case, and the shaft portion is inserted through the case-penetrating hole and the member-engaging hole; forming a fastening portion by crushing a tip end of the shaft portion and fastening the tip end of the shaft portion to a circumferential periphery of the member-engaging hole on an outside of the external terminal; and laser welding the fastening portion and the external terminal seamlessly along a plurality of weld lines that are set at two or more locations of the fastening portion that are circumferentially spaced from each other and that are set so as to extend across a periphery of the fastening portion in a radial direction along an outer surface of the fastening portion by continuously shifting a position to which a laser is applied along each of the weld lines, wherein, in the step of laser welding of each weld line, welded portion has a length of K1 along the weld line in the radial direction and a width of K2 along a direction perpendicular to the weld line, and the length K1 is at least 1.5 times longer than the width of K2, in the step of laser welding of each weld line, a region that is to be laser welded is set to be at least 2.5 times longer in a region external to the fastening portion than in a region internal to the fastening portion, and in the step of laser welding of each weld line, the welded portion does not extend into an inside of the member-engaging hole when viewed axially along the shaft portion.
 2. The method according to claim 1, wherein the welded portion is disposed on a top surface of the fastening portion and a top surface of the external terminal.
 3. The method according to claim 1, wherein the laser welding is performed while continuously shifting the position to which the laser is applied along the weld line such that no spot-shaped welding marks remain after laser welding.
 4. The method according to claim 1, wherein the plurality of welds are circumferentially spaced from each other at varying intervals.
 5. The method according to claim 1, wherein the plurality of welds are circumferentially spaced from each other in an asymmetrical pattern.
 6. The method according to claim 1, the fastening portion is an umbrella shape where a center portion of the fastening portion is raised from an edge portion of the fastening portion.
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